Chapter4 Lesson1 Semiconductors

Q1. Give classification of metals, conductors and semiconductors on the basis of conductivity
Q2. (i) What are elemental semiconductors, give some examples.
(ii) What are Compound semiconductors and discuss their types.
Q3. On the basis of energy bands define metals, semiconductors and insulators. Define Fermi energy level and energy band gap.
Q4.What are Intrinsic semiconductor? Discuss behavior of an intrinsic semiconductor at T = 0 K and at T > 0 K, represent both cases diagrammatically on energy level diagram. Discuss how thermally generated electron-hole pairs are responsible for current?
Q5. What do you mean by extrinsic semiconductor? Discuss formation of n-type semiconductor along with energy level diagram.
Q6. What do you mean by doping? Discuss formation of p-type semiconductor along with energy level diagram.
Q7. What changes occur in energy bands of (a) n-type semiconductor at T > 0K, (b) p-type semiconductor at T > 0K in comparison of pure semiconductors?
Q8. Discuss formation of p-n junction. Define depletion layer, barrier potential in a p-n junction
Q9. What is a semiconductor diode? Explain the symbol for p-n junction diode. Discuss V-I characteristics of semiconductor diode under forward bias and under reverse bias
Q10. Discuss energy level diagram of p-n junction across Barrier potential under reverse bias and forward bias (1) without battery, (2) Low battery voltage, and (3) High voltage battery.
Q11. What do you mean by threshold voltage or cut-in voltage? Give its value for germanium diode and for silicon diode.
Q12. Define reverse saturation current and break down voltage for p-n junction. Define a quantity called dynamic resistance for semiconductor diode.
Q13. Give application of junction diode as a half wave rectifier
Q14. What do you mean by centre-tap transformer? How it is used in full wave rectifier?
Q15. What do you mean by pulsating DC? How we can filter out the ac ripple?

Numerical

Q16. C, Si and Ge have same lattice structure. Why is C insulator while Si and Ge intrinsic semiconductors?
Q17. Can we take one slab of p-type semiconductor and physically join it to another n-type semiconductor to get p-n junction?
Q18. Suppose a pure Si crystal has 5×10^{28} atoms m^{-3}. It is doped by 1 ppm concentration of pentavalent As. Calculate the number of electrons and holes. Given that $n_i = 1.5 \times 10^{16}$ m^{-3}.
Q19. What is the behavior of semiconductors at room temperature? (1)
Q20. Find the hole density in doped silicon if at 27° C, both the electron density and the hole density are equal to $5x10^{16}$ m^{-3}. On doping with arsenic, electron density increases to 11×10^{19} m^{-3}. (2)
Q21. Draw the energy band diagrams of (*i*) *n*-type and (*ii*) *p*-type semiconductor at temperature, *T*> 0 K. In the case *n*-type Si semiconductor, the donor energy level is slightly below the bottom of conduction band, whereas in *p*-type semiconductor the acceptor energy level is slightly above the top of the valence band. Explain, what role do these energy levels play in conduction and valence bands. (3)

Q22. The V-I characteristic of a silicon diode is shown in the Fig. Calculate the resistance of the diode at
(a) I_D = 15 mA and
(b) V_D = –10 V.

